



CyberCareer

IT Security
Skill Advisor

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Helping IT Security Students Find Jobs & Close Skill Gaps

Analyse CV · Job Market Germany · Gap Analysis · AI Roadmap

4

AI Agents

6+

APIs Integrated

42

Unit Tests

100%

GDPR Compliant

The Problem

Why this tool exists

No visibility on required skills

Students don't know which skills are missing for IT Security jobs in Germany.

Fragmented job search

Indeed, StepStone, Bundesagentur — each platform separate, no unified tool.

No structured learning plan

Even when a gap is identified, no concrete roadmap to fill it.

Our Solution

- ✓ Upload CV (PDF / text)
- ✓ Auto skill detection
- ✓ Match with real job offers
- ✓ Visual gap analysis
- ✓ AI-powered learning plan

Multi-Agent Architecture

Inspired by the agent-skills framework — 4 specialized agents

01

Scout Agent

Job Discovery

CV upload · PDF parsing · Skill extraction · Multi-platform job search

Sprint 1 ✓

02

Matcher Agent

Gap Analysis

Matching score · Missing skills · CV vs job offers comparison in real-time

Sprint 1 ✓

03

Writer Agent

Letter Generation

Personalized cover letter generation using Claude AI

Sprint 2

04

Tracker Agent

Applications

Full CRUD application tracker · Status management · History

Sprint 1 ✓

Tech Stack

Deliberate technical choices — zero unnecessary dependencies

Backend	Node.js (no framework)	No heavy dependencies, immediate deployment, native HTTP server
Frontend	Vanilla JS + HTML/CSS	No build step needed, instant load, accessible without npm install
PDF Parser	pdf-parse (npm)	Handles UTF-16, CMap, Word/Canva/Adobe PDFs — server-side, no CDN
Auth	sCrypt (Node built-in)	Secure password hashing, no external lib — unique random salt
Storage	Local JSON (storage.json)	Simple persistence, human-readable, 100% local — GDPR by design
AI / LLM	Claude API (Anthropic)	Personalized roadmap, profile analysis, contextual suggestions

API Integrations — Job Search Germany

Free sources require no token — ready to use out of the box

Free — No token required

Bundesagentur für Arbeit

Official German job board — jobboerse-jobsuche API key

Arbeithnow

Public API — 300+ tech jobs per page, Germany-focused

Remotive

Remote IT jobs worldwide — JSON REST API

LinkedIn (guest)

Public endpoint, no authentication required

Optional — Token required

Apify — StepStone.de

\$1 / 1,000 results — professional scraping

Apify — Indeed.de

\$0.003 / listing — structured job data

Jooble

Aggregates 140k+ sites — free API key

ⓘ StepStone & Indeed provide no public read API — official APIs are reserved for ATS recruitment partners only. Solution: Apify scraper or Bundesagentur data.

Sprint 1 — Delivery

Foundation · May 2025 · All requirements met



Architecture design

Multi-agents: Scout / Matcher / Writer / Tracker



Tech stack setup

Node.js + Vanilla JS — zero build step required



Legal & GDPR research

Consent banner + full Datenschutzerklärung (Art. 6 GDPR)



CV Parser (PDF + text)

pdf-parse server-side — UTF-16, CMap, Word/Adobe/Canva



IT Security skills database

15 skills, 4 roles (SOC, Pentest, Cloud, Consultant)



Gap Analysis Engine

Score, matched/missing skills, suggestions per skill



Bundesagentur API

Free, no token + Arbeitnow + Remoteive + LinkedIn guest



Secure authentication

script + random salt, 24h tokens, register/login/logout



Application Tracker

Full CRUD: add, update status, delete applications



Unit Tests (42 tests)

node tests/test.js — no external framework required



README documentation

Install guide, API docs, GDPR, architecture, roadmap

Tests & Quality

42 unit tests — no external framework — node tests/test.js

42

tests passed
100% ✓

normalize()

3 tests



findSkills()

7 tests



analyzeRoles()

6 tests



hashPassword()

5 tests



decodePdfStr()

4 tests



stripHtml()

3 tests



kwTokens()

3 tests



matchesKeyword()

3 tests



jobMatchesProfile()

3 tests



Edge cases

5 tests



GDPR / DSGVO Compliance

Privacy by Design — 100% local, no data transmitted without consent

Art. 5

Data minimization

Only login token + applications stored — locally on user's machine, nothing else

Art. 15-17

User rights

Full access via storage.json · Deletion: rm storage.json · Export: readable JSON

Art. 32

Data security

script + unique random salt per password · Tokens expire after 24h automatically

Art. 6

Legal basis

Explicit consent (banner) + legitimate interest for technical session functions

Art. 25

Privacy by Design

No cloud, no tracking, passwords hashed with script — no plaintext ever stored

Art. 13

Transparency

Full Datenschutzerklärung integrated in the app as a modal (German + English)

Roadmap — Next Steps

DONE

Sprint 1

May 2025

- ✓ Multi-agent architecture
- ✓ PDF CV Parser
- ✓ Gap Analysis Engine
- ✓ Bundesagentur API
- ✓ Secure authentication
- ✓ 42 Unit tests
- ✓ GDPR compliance

UP NEXT

Sprint 2

June 2025

- Writer Agent (AI letters)
- 200+ security skills
- Jooble + Apify UI toggle
- Email notifications
- Advanced user profile

PLANNED

Sprint 3

July 2025

- PDF / Excel export
- Statistics dashboard
- Certification recommendations
- Mobile-friendly UI
- Public REST API

Thank You

Questions & Discussion

<https://github.com/Calderon112/cv-skill-advisor.git>

Jardel Kenne Tedjeu
Gelsenkirchen · May 2025

42 / 42

Tests passed

6

APIs integrated

GDPR

Compliant

Live demo available at localhost:3000

All 42 tests green · PDF bug fixed · GDPR banner implemented